

LABORATOR 2: Ecuatii differentiale

Initializare

```
> restart:                               sterge din memorie valori si variabile memorate anterior
> with(DEtools):                         incarca pachetul pt rezolvarea ecuatiilor differentiale
> with(plots):                           incarca pachetul de grafica
Warning, the name changecoords has been redefined
```

Operatia de derivare (recapitulare)

Pentru ecuatiile differentiale de ordin superior avem nevoie de definirea derivatelor de ordin superior. De exemplu sa consideram functia $f(x)=x^4+x^2+2$.

```
> restart:
> f:=x->x^4+x^2+2;
```

$$f := x \rightarrow x^4 + x^2 + 2$$

Derivata de ordinul 1 se calculeaza cu ajutorul comenzii **diff**

```
> diff(f(x),x);
```

$$4x^3 + 2x$$

Pentru derivatele de ordin superior se utilizeaza aceeasi comanda dar se pune variabila de mai multe ori, de exemplu pt derivata de ordinul 2 avem **diff(f(x),x,x)**

```
> diff(f(x),x,x);
```

$$12x^2 + 2$$

In cazul in care dorim calculul derivatei de ordinul 4, putem proceda ca mai inainte:

```
diff(f(x),x,x,x,x) punand varibila de 4 ori sau se poate simplifica scrierea utilizand comanda:
diff(f(x),x$4)
> diff(f(x),x$4);
```

$$24$$

O alta modalitate de a calcula derivata este prin utilizarea operatorului de derivare **D**

```
> D(f)(x);
```

$$4x^3 + 2x$$

Operatorul este utilizat atunci cind avem nevoie de valoare derivatei intr-un anumit punct si este folosit pentru precizarea conditiilor initiale

```
> D(f)(0);
```

$$0$$

Pentru derivari de ordin superior se utilizeaza compunerea operatorului de derivare, de exemplu pentru derivata de ordinul 2 avem **(D@D)(f)(x)** sau **(D@@2)(f)(x)**. Pentru derivata de ordinul 3 avem **(D@D@D)(f)(x)** sau **(D@@3)(f)(x)**

```
> (D@D)(f)(x);
```

$$12x^2 + 2$$

```
> (D@D)(f)(2);
```

50

> (D@@2)(f)(x);

$12x^2 + 2$

> (D@@4)(f)(x);

24

Definirea si rezolvarea unei ecuatii diferentiale

Fie ecuatia diferentiala de ordinul 1: $y'(x) = \sin(x) * (y(x))^2$. Aceasta ecuatie este o ecuatie cu variabile separabile adica este forma $y'(x) = f(x) * g(y)$. Ecuatia se defineste in MAPLE utilizand comanda **diff** dupa cum urmeaza:

> **ecdif1:=diff(y(x),x) =sin(x)*(y(x))^2;**

$$ecdif1 := \frac{d}{dx} y(x) = \sin(x) y(x)^2$$

Pentru a obtine solutia generala se utilizeaza comanda **dsolve(ecuatie, functie necunoscuta)**

> **dsolve(ecdif1,y(x));**

$$y(x) = \frac{1}{\cos(x) + _C1}$$

Metodele incercate si utilizate de **MAPLE** pentru a obtine solutiile ecuatiilor diferentiale pot fi observate crescand **infolevel** pentru **dsolve** la 3:

> **infolevel[dsolve]:=3;**

$infolevel_{dsolve} := 3$

apoi reexecutam comanda **dsolve**:

> **dsolve(ecdif1,y(x));**

Methods for first order ODEs:
--- Trying classification methods ---
trying a quadrature
trying 1st order linear
trying Bernoulli
<- Bernoulli successful

$$y(x) = \frac{1}{\cos(x) + _C1}$$

In cazul acestei ecuatii comanda **dsolve** a sesizat ca aceasta ecuatie este de tip Bernoulli, adica este forma $y'(x) + P(x) * y(x) = Q(x) * (y(x))^{\alpha}$ unde **alpha** este diferit de 0 si 1. Daca dorim, putem cere in comanda **dsolve** sa se aplice metoda rezolvarii ecuatiilor separabile prin specificarea acestei optiuni dupa cum urmeaza

> **dsolve(ecdif1,y(x),[separable]);**

Classification methods on request
Methods to be used are: [separable]

* Tackling ODE using method: separable
--- Trying classification methods ---
trying separable
<- separable successful

$$y(x) = -\frac{1}{-\cos(x) + _C1}$$

Pentru a vedea care sunt metodele de rezolvare a ecuatiilor de ordinul 1 implementate in **dsolve** se poate da comanda:

```
> `dsolve/methods`[1];
[quadrature, linear, Bernoulli, separable, inverse_linear, homogeneous, Chini, lin_sym,
 exact, Abel, pot_sym]
```

Pentru alte amanunte legate de comanda **dsolve** executati:

```
> ?dsolve;
```

Pentru suprimarea informatiilor suplimentare resetam **infolevel** pentru **dsolve** la 0

```
> infolevel[dsolve]:=0;
infoleveldsolve := 0
```

In unele cazuri este mai convenabil obtinerea solutiilor in forma implicita, acest lucru se poate realiza specificand in cadrul procedurii **dsolve** optiunea **implicit**. De exemplu, sa consideram ecuatia diferentiala $(3y^2 + e^x) y' + e^x (y + 1) + \cos x = 0$:

```
> ecdif2:=(3*y(x)^2+exp(x))*diff(y(x),x)+exp(x)*(y(x)+1)+cos(x) = 0;
```

$$ecdif2 := (3 y(x)^2 + e^x) \left(\frac{d}{dx} y(x) \right) + e^x (y(x) + 1) + \cos(x) = 0$$

```
> dsolve(ecdif2,y(x),implicit);
```

$$e^x y(x) + e^x + \sin(x) + y(x)^3 + _C1 = 0$$

Pentru a vedea avantajul, incercati sa rezolvati ecuatia fara a preciza optiunea **implicit**.

Pentru ecuatiile diferentiale de ordinul 2 se foloseste aceeasi metoda, se defineste ecuatia si apoi se utilizeaza comanda **dsolve**. De exemplu sa consideram ecuatia $y''(x) + 3y'(x) + 2y(x) = 1 + x^2$

```
> ecdif3:=diff(y(x),x$2)+3*diff(y(x),x)+2*y(x)=1+x^2;
```

$$ecdif3 := \left(\frac{d^2}{dx^2} y(x) \right) + 3 \left(\frac{d}{dx} y(x) \right) + 2 y(x) = 1 + x^2$$

```
> dsolve(ecdif3,y(x));
```

$$y(x) = \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - e^{(-2x)} _C1 + e^{(-x)} _C2$$

Reprezentarea grafica a solutiilor

```
> with(plots):
```

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Sa consideram, din nou, prima ecuatie: $y'(x) = \sin(x) * (y(x))^2$.

```
> ecdif1:=diff(y(x),x) =sin(x)*(y(x))^2;
```

$$ecdif1 := \frac{d}{dx} y(x) = \sin(x) y(x)^2$$

```
> sol1:=dsolve(ecdif1,y(x));
```

$$sol1 := y(x) = \frac{1}{\cos(x) + _C1}$$

Rezultatul comenzii **dsolve** nu este o functie, ci este o ecuatie. Pentru manipularea solutiei exista doua alternative, fie definim functia ce reprezinta solutia (in situatia in care expresia ei nu e prea complicata), in cazul dat solutia depinde de variabila independenta x si constanta de integrare:

> **y1:=(x,c)->1/(cos(x)+c);**

$$y1 := (x, c) \rightarrow \frac{1}{\cos(x) + c}$$

sau avem acces la membrul drept utilizand comanda **rhs** (*right hand side*) si apoi comanda **unapply** pentru a construi solutia ca functie

> **right_hand_expr:=rhs(sol1);**

$$right_hand_expr := \frac{1}{\cos(x) + _C1}$$

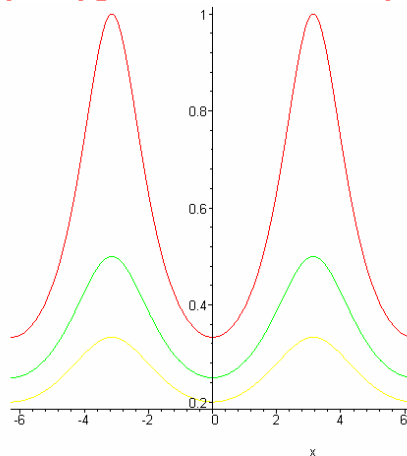
Prin comanda **unapply** se transforma expresia **right_hand_expr** in functie precizand variabilele acesteia:

> **y2:=unapply(right_hand_expr,x,_C1);**

$$y2 := (x, _C1) \rightarrow \frac{1}{\cos(x) + _C1}$$

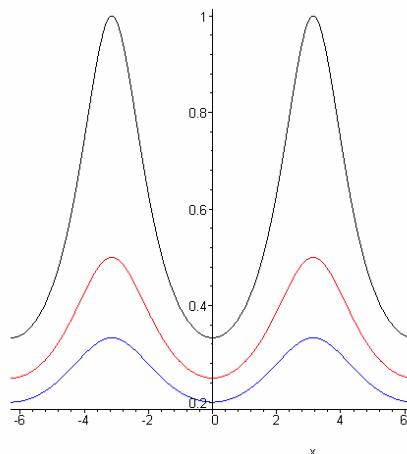
Pentru reprezentarea grafica a catorva solutii dam valori constantei **c**, de exemplu **c:=2**, **c:=3** si **c:=4**

> **plot([y1(x,2),y1(x,3),y1(x,4)],x=-2*Pi..2*Pi);**



Daca se doreste obtinerea graficelor cu anumite culori precizate vom utiliza urmatoarea comanda:

> **plot([y1(x,2),y1(x,3),y1(x,4)],x=-2*Pi..2*Pi,color=[black,red,blue]);**



In cazul celei de a doua ecuatii, $(3y^2 + e^x) y' + e^x (y + 1) + \cos x = 0$, unde solutia este obtinuta in forma implicita, trebuie utilizata procedura **implicitplot**.

> **ecdif2:=(3*y(x)^2+exp(x))*diff(y(x),x)+exp(x)*(y(x)+1)+cos(x) = 0;**

$$ecdif2 := (3 y(x)^2 + e^x) \left(\frac{d}{dx} y(x) \right) + e^x (y(x) + 1) + \cos(x) = 0$$

> **sol2:=dsolve(ecdif2,y(x),implicit);**

$$sol2 := e^x y(x) + e^x + \sin(x) + y(x)^3 + _C1 = 0$$

Construim functia care ne da ecuatia implicita a solutiilor, in acest caz, expresia se afla in partea stanga a ecuatiei si vom folosi comanda **lhs** (left hand side) pentru a avea acces la aceasta expresie

> **left_hand_side:=lhs(sol2);**

$$left_hand_side := e^x y(x) + e^x + \sin(x) + y(x)^3 + _C1$$

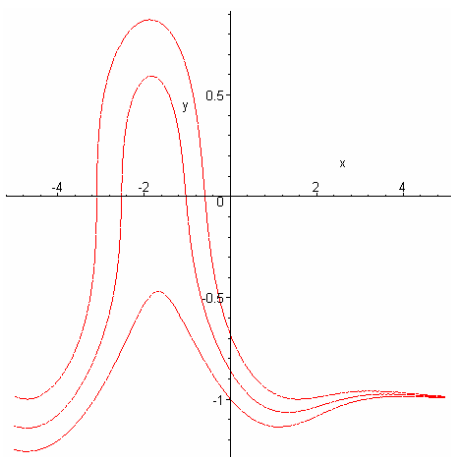
> **lhs1:=subs(y(x)=y, left_hand_side);**

$$lhs1 := e^x y + e^x + \sin(x) + y^3 + _C1$$

> **f:=unapply(lhs1,x,y,_C1);**

$$f := (x, y, _C1) \rightarrow e^x y + e^x + \sin(x) + y^3 + _C1$$

> **implicitplot({f(x,y,0)=0,f(x,y,0.5)=0,f(x,y,1)=0},x=-5..5,y=-5..5,numpoints=10000);**



Daca dorim reprezentarea grafica a mai multor solutii, putem genera un sir de functii $f(x,y,c)$ pentru $c = -4, -19/5, \dots, -1/5, 0, 1/5, 2/5, \dots, 4$ folosim comanda **seq** dupa cum urmeaza

> **sir_sol:=seq(f(x,y,i/5)=0,i=-20..20);**

$$sir_sol := e^x y + e^x + \sin(x) + y^3 - 4 = 0, e^x y + e^x + \sin(x) + y^3 - \frac{19}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{18}{5} = 0, e^x y + e^x + \sin(x) + y^3 - \frac{17}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{16}{5} = 0, e^x y + e^x + \sin(x) + y^3 - 3 = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{14}{5} = 0, e^x y + e^x + \sin(x) + y^3 - \frac{13}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{12}{5} = 0, e^x y + e^x + \sin(x) + y^3 - \frac{11}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - 2 = 0, e^x y + e^x + \sin(x) + y^3 - \frac{9}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{8}{5} = 0, e^x y + e^x + \sin(x) + y^3 - \frac{7}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{6}{5} = 0, e^x y + e^x + \sin(x) + y^3 - 1 = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{4}{5} = 0, e^x y + e^x + \sin(x) + y^3 - \frac{3}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 - \frac{2}{5} = 0, e^x y + e^x + \sin(x) + y^3 - \frac{1}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 = 0, e^x y + e^x + \sin(x) + y^3 + \frac{1}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + \frac{2}{5} = 0, e^x y + e^x + \sin(x) + y^3 + \frac{3}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + \frac{4}{5} = 0, e^x y + e^x + \sin(x) + y^3 + 1 = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + \frac{6}{5} = 0, e^x y + e^x + \sin(x) + y^3 + \frac{7}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + \frac{8}{5} = 0, e^x y + e^x + \sin(x) + y^3 + \frac{9}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + 2 = 0, e^x y + e^x + \sin(x) + y^3 + \frac{11}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + \frac{12}{5} = 0, e^x y + e^x + \sin(x) + y^3 + \frac{13}{5} = 0,$$

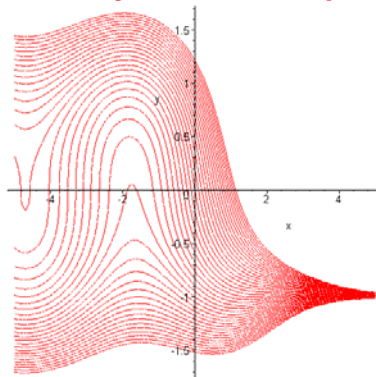
$$e^x y + e^x + \sin(x) + y^3 + \frac{14}{5} = 0, e^x y + e^x + \sin(x) + y^3 + 3 = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + \frac{16}{5} = 0, e^x y + e^x + \sin(x) + y^3 + \frac{17}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + \frac{18}{5} = 0, e^x y + e^x + \sin(x) + y^3 + \frac{19}{5} = 0,$$

$$e^x y + e^x + \sin(x) + y^3 + 4 = 0$$

> **implicitplot({sir_sol},x=-5..5,y=-5..5,numpoints=10000);**



In cazul ecuatiei de ordinul 2, $y''(x) + 3y'(x) + 2y(x) = 1 + x^2$, reprezentarea grafica a solutiilor revine la particularizarea celor doua constante de integrare

> **ecdif3:=diff(y(x), x\$2)+3*diff(y(x), x)+2*y(x)=1+x^2;**

$$ecdif3 := \left(\frac{d^2}{dx^2} y(x) \right) + 3 \left(\frac{d}{dx} y(x) \right) + 2 y(x) = 1 + x^2$$

> **sol3:=dsolve(ecdif3, y(x));**

$$sol3 := y(x) = \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - e^{(-2x)}_C1 + e^{(-x)}_C2$$

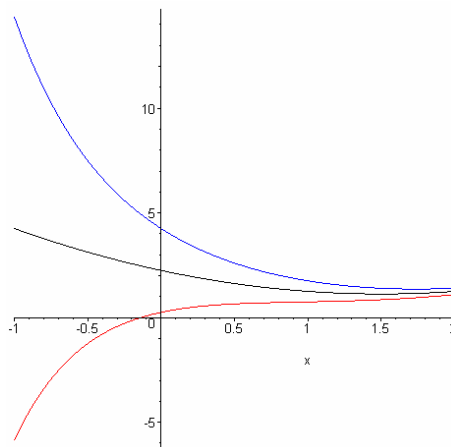
> **right_hand_expr:=rhs(sol3);**

$$right_hand_expr := \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - e^{(-2x)}_C1 + e^{(-x)}_C2$$

> **y1:=unapply(right_hand_expr, x, _C1, _C2);**

$$y1 := (x, _C1, _C2) \rightarrow \frac{1}{2}x^2 - \frac{3}{2}x + \frac{9}{4} - e^{(-2x)}_C1 + e^{(-x)}_C2$$

> **plot([y1(x, 0, 0), y1(x, -1, 1), y1(x, 1, -1)], x=-1..2, color=[black, blue, red]);**



sau putem reprezenta un sir de solutii:

> **sir_sol3:=seq(seq(y1(x, i/5, j/2), i=-2..2), j=-2..2);**

$$sir_sol3 := \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{2}{5}e^{(-2x)} - e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{1}{5}e^{(-2x)} - e^{(-x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{5}e^{(-2x)} - e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{2}{5}e^{(-2x)} - e^{(-x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{2}{5}e^{(-2x)} - \frac{1}{2}e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{1}{5}e^{(-2x)} - \frac{1}{2}e^{(-x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{2}e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{5}e^{(-2x)} - \frac{1}{2}e^{(-x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{2}{5}e^{(-2x)} - \frac{1}{2}e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{2}{5}e^{(-2x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{1}{5}e^{(-2x)},$$

$$\frac{1}{2}x^2 - \frac{3}{2}x + \frac{9}{4}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{5}e^{(-2x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{2}{5}e^{(-2x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{2}{5}e^{(-2x)} + \frac{1}{2}e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{1}{5}e^{(-2x)} + \frac{1}{2}e^{(-x)},$$

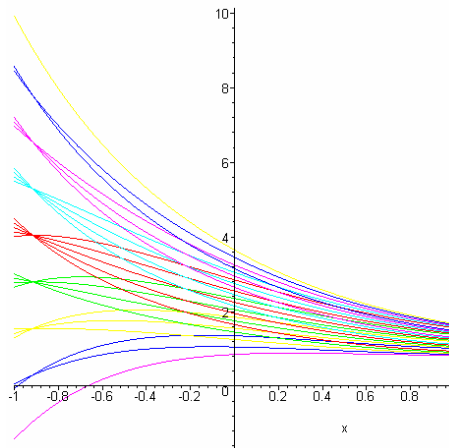
$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{1}{2}e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{5}e^{(-2x)} + \frac{1}{2}e^{(-x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{2}{5}e^{(-2x)} + \frac{1}{2}e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{2}{5}e^{(-2x)} + e^{(-x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + \frac{1}{5}e^{(-2x)} + e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} + e^{(-x)}, \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{5}e^{(-2x)} + e^{(-x)},$$

$$\frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{2}{5}e^{(-2x)} + e^{(-x)}$$

> **plot([sir_sol3], x=-1..1);**



Rezolvarea problemelor cu valori initiale (Probleme Cauchy)

În general, rezolvarea anumitor probleme revin la determinarea unei soluții pentru o ecuație diferențială ce satisface anumite condiții initiale. Aceste probleme se numesc probleme cu valori initiale sau probleme Cauchy. De exemplu, să presupunem că trebuie determinată soluția ecuației $y'(x) = \sin(x) * (y(x))^2$ ce satisface condiția $y(0) = 1/3$, adică soluția reprezentată în paragraful anterior pentru constanta $c=2$

> **restart:with(DEtools):**

> **ecdif1:=diff(y(x),x) =sin(x)*(y(x))^2;**

$$ecdif1 := \frac{d}{dx} y(x) = \sin(x) y(x)^2$$

Definim condiția inițială:

> **cond_in:=y(0)=1/3;**

$$cond_in := y(0) = \frac{1}{3}$$

Comanda de rezolvare a problemei Cauchy este similară cu cea de rezolvare a ecuației la care se adaugă și condiția inițială:

> **sol1:=dsolve({ecdif1,cond_in},y(x));**

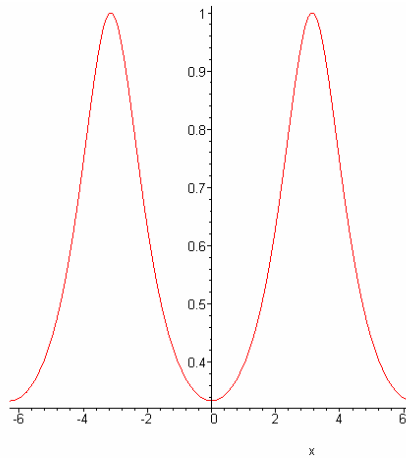
$$sol1 := y(x) = \frac{1}{\cos(x) + 2}$$

Pentru reprezentarea grafică a soluției se utilizează comanda **rhs**:

> **y1:=unapply(rhs(sol1),x);**

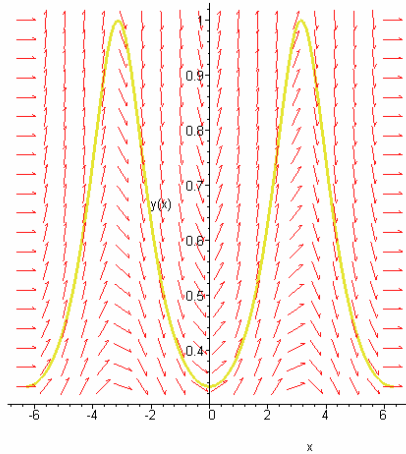
$$y1 := x \rightarrow \frac{1}{\cos(x) + 2}$$

> **plot(y1(x), x=-2*Pi..2*Pi);**



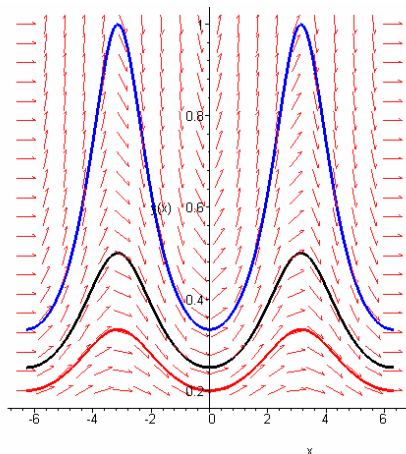
Se poate obtine graficul solutiei problemei Cauchy utilizand comanda **DEplot**:

> **DEplot(ecdif1, y(x), x=-2*Pi..2*Pi, [[cond_in]], stepsize=0.1);**



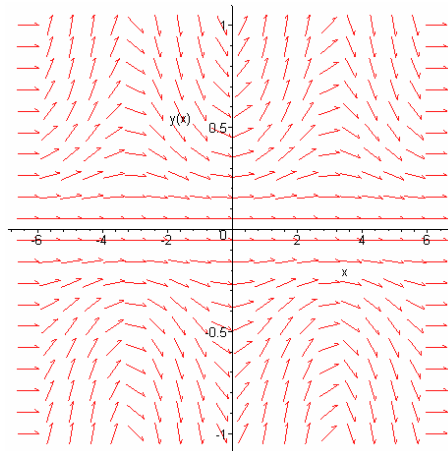
Se observa ca in aceasta reprezentare apare si campul de directii impreuna cu solutia. Daca se doreste reprezentarea grafica a solutiilor pentru diverse conditii initiale, de exemplu **y(0)=1/3**, **y(0)=1/4**, **y(0)=1/5**, se utilizeaza aceeasi comanda specificand lista de conditii initiale:

> **DEplot(ecdif1, y(x), x=-2*Pi..2*Pi, [[y(0)=1/3], [y(0)=1/4], [y(0)=1/5]], stepsize=0.1, linecolor=[black, red, blue]);**



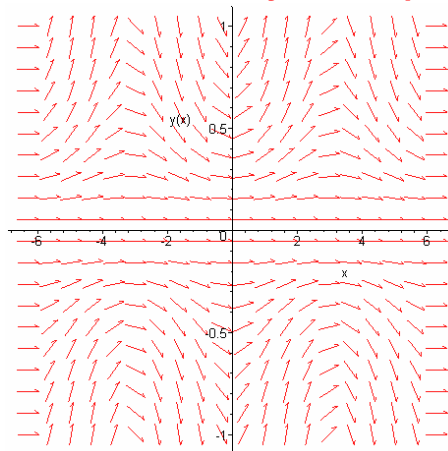
Daca dorim reprezentarea grafica doar a campului de directii se utilizeaza comanda:

> **DEplot(ecdif1,y(x),x=-2*Pi..2*Pi,y=-1..1);**



Acelasi rezultat se obtine utilizand si comanda **dfieldplot**:

> **dfieldplot(ecdif1,y(x),x=-2*Pi..2*Pi,y=-1..1);**



In cazul ecuatiilor diferentiale de ordinul 2 problema Cauchy va avea doua conditii initiale **$y(x_0)=a$** si **$y'(x_0)=b$** . Definirea celei de a doua conditii se face cu ajutorul operatorului de derivare **D**. De exemplu, sa determinam solutia ecuatiei **$y''(x)+3y'(x)+2y(x)=1+x^2$** ce satisface conditiile initiale **$y(0)=0$** si **$y'(0)=1$**

> **ecdif3:=diff(y(x),x\$2)+3*diff(y(x),x)+2*y(x)=1+x^2;**

$$ecdif3 := \left(\frac{d^2}{dx^2} y(x) \right) + 3 \left(\frac{d}{dx} y(x) \right) + 2 y(x) = 1 + x^2$$

> **cond_in:=y(0)=0,D(y)(0)=1;**

$$cond_in := y(0) = 0, D(y)(0) = 1$$

> **sol3:=dsolve({ecdif3,cond_in},y(x));**

$$sol3 := y(x) = \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{4}e^{(-2x)} - 2e^{(-x)}$$

Pentru reprezentarea grafica a solutiei fie utilizam metoda de constructie a solutiei cu **rhs** si **unapply** sau, direct, prin **DEplot**

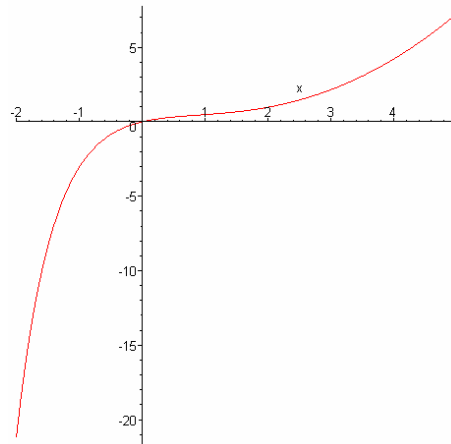
> **rhs3:=rhs(sol3);**

$$rhs3 := \frac{x^2}{2} - \frac{3x}{2} + \frac{9}{4} - \frac{1}{4}e^{(-2x)} - 2e^{(-x)}$$

> **y3:=unapply(rhs3,x);**

$$y3 := x \rightarrow \frac{1}{2}x^2 - \frac{3}{2}x + \frac{9}{4} - \frac{1}{4}e^{(-2x)} - 2e^{(-x)}$$

> **plot(y3(x),x=-2..5);**



> **DEplot(ecdif3,y(x),x=-2..5,[[cond_in]],stepsize=0.1);**

